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Thermodynamic Assessments of the Au-Nd and Au-Dy Systems

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The thermodynamic assessments of the Au-Nd and Au-Dy binary systems have been carried out by the Calculation of Phase Diagram (CALPHAD) method based on the available experimental data. The Gibbs free energies of the solution phases including liquid, fcc, bcc, hcp and dhcp were described by the substitutional solution models with the Redlich-Kister equation, and those of the intermetallic compounds were described by the sublattice models. A set of self-consistent and reasonable thermodynamic parameters is obtained for the binary systems, which describes the Gibbs free energies of the solution phases and the intermetallic compounds phases.

Keywords CALPHAD, phase diagram, thermodynamic assessment, thermodynamic modeling

1. Introduction

In the fabrication devices, the reliability of integrated circuit and Micro-Electro-Mechanical Systems devices packaging depends on the soldering systems to a great extent.[1,2] Owing to the excellent heat conductivity, electrical conductivity, thermostability and outstanding corrosion stability mechanical properties, Sn-Au-based filler alloy is treated as the important low-temperature lead-free solders in electronic devices.[3,4] The addition of rare earth (RE) elements can control and eliminate the negative effect of impurity elements, which can provide a direct and powerful connection on these surfaces of Sn-Au-based solder.[5-7] Therefore, to design the appropriate solders compositions in the Sn-Au-RE system effectively, understanding of the phase equilibria becomes necessary. In the research program, the thermodynamic database for Au-RE (RE = La, Ce, Pr, Nd, Eu, Gd, Dy, Er, Ho….) binary systems will be developed. Here we present the thermodynamic assessment of the Au-Nd and Au-Dy binary systems. Next step, we will optimize the other Au-RE binary systems.

Recently, the Au-Nd binary system was thermodynamically optimized in Ref. 8, where the enthalpies of mixing of liquid phase[9] was not considered. On the basis of Ref. 8, we reassessed the thermodynamic parameters for the phase diagrams in the Au-Nd system. In the present work, the thermodynamic assessments of the Au-Nd and Au-Dy binary systems have been carried out by the CALPHAD approach,[10,11] which will provide important thermodynamic information for the multicomponent Sn-Au-RE alloy systems.

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2. Thermodynamic Models

2.1 Liquid Phases and Solid Solutions

The Gibbs free energies of the liquid, fcc, bcc, hcp, and dhcp phases in the Au-RE (RE = Nd, Dy) binary systems are described by the substitutional solution model with using the Redlich-Kister formula.[12] The molar Gibbs free energies of solution phase in the Au-RE systems are expressed as:

$$G_m^{\Phi} = \sum_{i=\mathrm{Au,RE}} {}^0G_i^{\Phi} x_i + RT \sum_{i=\mathrm{Au,RE}} x_i \ln x_i + x_{\mathrm{Au}} x_{\mathrm{RE}} \sum_{m=0}^{n} {}^mL_{\mathrm{Au,RE}}^{\Phi}(x_{\mathrm{Au}} - x_{\mathrm{RE}}). \quad \text{(Eq 1)}$$

where ${}^0G_i^{\Phi}$is the Gibbs free energy of pure element i in the each reference state, R is the gas constant, T is the temperature, x_i is the molar Gibbs energy of component i. ${}^mL_{\mathrm{Au,RE}}^{\Phi}$ is the binary interaction parameter and formulated as:

$${}^mL_{\mathrm{Au,RE}}^{\Phi} = a_m + b_m T. \quad \text{(Eq 2)}$$

where the values of a_m and b_m are optimized in the present work.

2.2 Stoichiometric Compounds

In the Au-RE (RE = Nd, Dy) binary systems, the intermetallic compounds are treated as stoichiometric compounds. The Gibbs free energy of the Au_mRE_n compound is described by the sublattice model,[13] which is expressed as:

$$\Delta^0 G_f^{Au_m RE_n} = {}^0 G_f^{Au_m RE_n} - m^0 G_{Au}^{SER} - n^0 G_{RE}^{SER} = a'_m + b'_m T + c'_m T \ln T + d'_m T^2 + e'_m T^3 \quad \text{(Eq 3)}$$

where the ${}^0G_f^{Au_m RE_n}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements. The parameters of a'_m, b'_m, c'_m, , d'_m, and e'_m are assessed in the present work.

3. Thermodynamic Optimization

The optimization of the thermodynamic parameters was carried out by using PARROT program in the Thermo-Calc software,[14] which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the available experimental data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits

3.1 The Au-Nd system

3.1.1 Experimental Information. In the Au-Nd system, eight intermetallic compounds (Au_6Nd,[15] $Au_{51}Nd_{14}$,[16,17] $Au_{36}Nd_{17}$,[18] Au_4Nd_3,[18] AuNd (γ, β and α allotropic forms)[16,19] and $AuNd_2$[16] phases) were identified, respectively. In 1971, McMasters et al.[16] determined the melting and transformation temperatures of the AuNd phase by the both powder- and single-crystal x-ray diffraction (XRD) methods. Using the x-ray parametric, the terminal solid solubility of Nd in (Au) was obtained by Rider et al.[20] With differential thermal analysis (DTA), XRD, optical microscopy, scanning electron microscopy, and electron probe microanalysis, the phase equilibria of the Au-Nd system was studied by Saccone et al.,[21] who reported four solution phases (liquid, (Au), (αNd), (βNd)) and eight intermetallic compounds (Au_6Nd, $Au_{51}Nd_{14}$, $Au_{36}Nd_{17}$, Au_4Nd_3, γAuNd, βAuNd, αAuNd and $AuNd_2$ phases) in the Au-Nd binary system, and also determined four eutectic reactions (L ↔ (Au) + Au_6Nd at 9 at.% Nd and 795 °C, L ↔ $Au_{51}Nd_{14}$ + $Au_{36}Nd_{17}$ at 28 at.% Nd and 1050 °C, L ↔ $Au_{36}Nd_{17}$ + Au_4Nd_3 at 37 at.% Nd and 1080 °C, and L ↔ $AuNd_2$ + (αNd) at 81 at.% Nd and 655 °C) and three peritectic reactions (L + $Au_{51}Nd_{14}$ ↔ Au_6Nd at 875 °C, L + βAuNd ↔ Au_3Nd_4 at 1250 °C, and L + βAuNd ↔ $AuNd_2$ at 810 °C), two polymorphic reactions (the γAuNd ↔ βAuNd at 1360 °C and βAuNd ↔ αAuNd at 430 °C) and a catatectic reaction : (βNd) ↔ L + (αNd) at 825 °C.

In addition, the standard enthalpies of formation of the $Au_{51}Nd_{14}$, $Au_{36}Nd_{17}$ and αAuNd compounds, and enthalpies of mixing of liquid phase at 1473 ± 2 K were experimentally measured by Fitzner and Kleppa.[9]

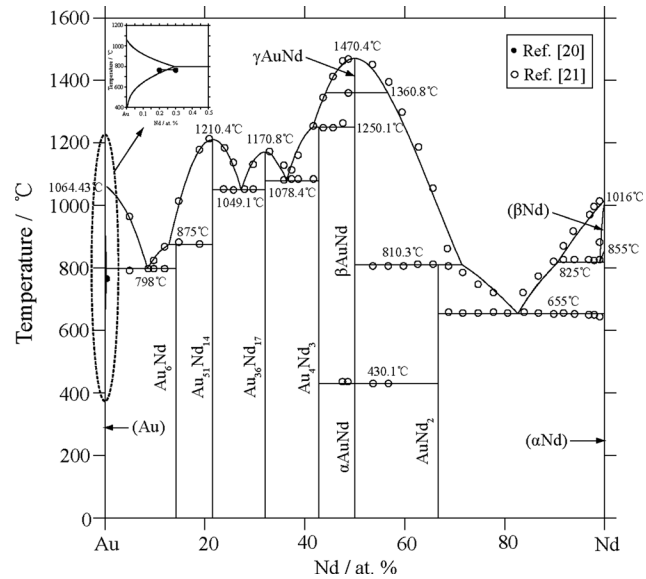

Fig. 1 Calculated phase diagram of the Au-Nd system compared with the experimental data[20,21]

3.1.2 Optimized Results and Discussion. The calculated Au-Nd phase diagram with the experimental data[20,21] is shown in Fig. 1. Table 1 presents the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data.[20,21] It is seen that the calculated results are in good agreement with the experimental data.[20,21] A complete set of self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Au-Nd system are given in Table 2.

As shown in Fig. 2, the calculated standard enthalpies of formation with reference states of Au (fcc) and αNd (dhcp) phases compared with the experimental data[9] are observed and considered acceptable. Moreover, the calculated mixing enthalpies of liquid phase at 1473 K with reference states of pure elements of Au (liquid) and Nd (liquid) in the Au-Nd system are illustrated in Fig. 3. It is seen that an agreement is obtained between the calculated results and the experimental data.[9]

3.2 The Au-Dy System

3.2.1 Experimental Information. In the Au-Dy system, the maximum solubility of Dy in (Au) was experimentally determined to be 2.3% at.% at 808 °C by Rider et al.[20] Using x-ray powder diffractometry, Sadagopan et al.[22] identified the Au_2Dy and Au_3Dy phases. Later, based on microscopic, DTA and XRD, McMasters and Gschneidner[23] investigated the Au-Dy phase diagram, and reported three stoichiometry compounds of the AuDy (melting point, 1660 °C), Au_2Dy (melting point, 1320 °C) and Au_3Dy (melting point, 1230 °C). McMasters and McMasters[23] also determined four eutectic reactions (L ↔ (Au) + Au_6Dy at 9.5 at.% Dy and 808 °C, L ↔ Au_3Dy + Au_2Dy at 27.0 at.% Dy and 1185 °C, L ↔ Au_2Dy + AuDy at 38.0 at.% Dy and 1235 °C and L ↔ $AuDy_2$ + (αDy) at

Table 1 Comparison between the calculated and experimental results[20,21] in the Au-Nd system

Invariant reaction	Reaction type	Composition at.% Nd			Temperature, °C	Reference
$\beta Nd \leftrightarrow L + \alpha Nd$	Catatectic	99.0	90.0	≥99.5	825	[21]
		99.1	90.9	99.8	825	This work
$L \leftrightarrow \alpha Nd + AuNd_2$	Eutectic	81.0	≥99.5	66.7	655	[21]
		82.6	99.8	66.7	655	This work
$L + \beta AuNd \leftrightarrow AuNd_2$	Peritectic	70.0	50.0	66.7	810	[21]
		71.5	50.0	66.7	810.3	This work
$L \leftrightarrow \gamma AuNd$	Congruent		50.0		1470	[21]
			50.0		1470.4	This work
$\gamma AuNd \leftrightarrow \beta AuNd$	Allotropic		50.0		1360	[21]
			50.0		1360.8	This work
$\beta AuNd \leftrightarrow \alpha AuNd$	Allotropic		50.0		430	[21]
			50.0		430.1	This work
$L + \beta AuNd \leftrightarrow Au_4Nd_3$	Peritectic	41.5	50.0	42.9	1250	[21]
		41.7	50.0	42.9	1250.1	This work
$L \leftrightarrow Au_4Nd_3 + Au_{36}Nd_{17}$	Eutectic	37.0	42.9	32.1	1080	[21]
		36.4	42.9	32.1	1078.4	This work
$L \leftrightarrow Au_{36}Nd_{17}$	Congruent		32.1		1170	[21]
			32.1		1170.8	This work
$L \leftrightarrow Au_{36}Nd_{17} + Au_{51}Nd_{14}$	Eutectic	28.0	32.1	21.5	1050	[21]
		27.3	32.1	21.5	1049.1	This work
$L \leftrightarrow Au_{51}Nd_{14}$	Congruent		21.5		1210	[21]
			21.5		1210.4	This work
$L + Au_{51}Nd_{14} \leftrightarrow Au_6Nd$	Peritectic	13	21.5	14.3	875	[21]
		12.9	21.5	14.3	875	This work
$L \leftrightarrow Au_6Nd + (Au)$	Eutectic	···	···	0.2-0.3	795	[20]
		9	14.3	0.3	795	[21]
		8.5	14.3	0.3	798	This work

79.0 at.% Dy and 935 °C), three peritectic reactions ($L + Au_{51}Dy_{14} \leftrightarrow Au_6Dy$ at 845 °C, $L + Au_3Dy \leftrightarrow Au_{51}Dy_{14}$ at 1095 °C, and $L + \alpha AuDy \leftrightarrow AuDy_2$ at 1065 °C) and a polymorphic reaction ($\beta AuDy \leftrightarrow \alpha AuDy$ at 1360 °C).

Additionally, Meschel and Kleppa[24] experimentally measured the standard enthalpies of formation of αAuDy, Au_2Dy and Au_3Dy phases by the high temperature direct synthesis calorimetry. De Boer and Pettifor[25] predicted the value of the enthalpy of mixing of liquid phase at 50 at.% Dy in the binary system is 74 kJ/mol; however, no temperature dependence was given in the work of de Boer and Pettifor.[25]

3.2.2 Optimized Results and Discussion. The calculated Au-Dy phase diagram compared with the experimental data[20,23] is presented in Fig. 4, and Table 3 lists the compositions of the phases and the temperatures of all the invariant equilibria. It is seen that the good agreement is obtained between the calculated results and most experimental data except for the temperature of the eutectic reaction ($L \leftrightarrow Au_2Dy + Au_3Dy$), where the discrepancy between the present evaluation and the proposed experimental information[23] is 24 °C. Okamoto[26,27] mentioned that the width on one side is either one half or double that on the other side, the asymmetry may be considered clearly unusual. With regard to Au_3Dy phase, the ratio is significantly more two times in the experimental phase diagram, consequently it should be modified. Accordingly, the calculated temperature of this eutectic reaction has to be sacrificed in order to adjust to the liquidus symmetry and the enthalpy of formation of the Au_3Dy. On the other hand, Okamoto[26,27] also indicated that a compound with sharp liquids will decompose eutectoidally at low temperature. Since no experimental information indicates that Au_3Dy phase decomposes, the reported liquidus should be flatter. Therefore, the discrepancy between the calculated temperature of the eutectic reaction ($L \leftrightarrow Au_2Dy + Au_3Dy$) and experimental data can be acceptable. Table 4 lists a complete set of self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Au-Dy system.

In addition, the calculated standard enthalpies of formation with reference states of Au (fcc) and αDy (hcp) phases compared with the experimental data[24] are shown in Fig. 5 and a comparison of mixing enthalpies at 1473 K with reference states of pure elements of Au (liquid) and Dy (liquid) phases between the calculated and predicted data[25]

Table 2 Thermodynamic parameters in the Au-Nd systems assessed in the present work

Parameters in each phase, J/mol
Liquid phase, format (Au, Nd)
${}^{0}L^{Liq}_{Au,Nd} = -273000 + 6.023T$
${}^{1}L^{Liq}_{Au,Nd} = -112900 - 2.863T$
${}^{2}L^{Liq}_{Au,Nd} = 27856 + 6.853T$
${}^{3}L^{Liq}_{Au,Nd} = 57068 - 9.6T$
Fcc phase, format (Au, Nd)
${}^{1}L^{Fcc}_{Au,Nd} = -239950$
Bcc phase, format (Au, Nd)
${}^{0}L^{Bcc}_{Au,Nd} = -137714$
Dhcp phase, format (Au, Nd)
${}^{0}L^{Dhcp}_{Au,Nd} = -122000$
Au_6Nd phase, format $(Au)_{0.857}(Nd)_{0.143}$
${}^{0}G^{Au_6Nd}_{Au:Nd} - 0.857{}^{0}G^{fcc}_{Au} - 0.143{}^{0}G^{dhcp}_{Nd} = -44300 + 3.01T$
$Au_{51}Nd_{14}$ phase, format $(Au)_{0.785}(Nd)_{0.215}$
${}^{0}G^{Au_{51}Nd_{14}}_{Au:Nd} - 0.785{}^{0}G^{fcc}_{Au} - 0.215{}^{0}G^{dhcp}_{Nd} = -55780 - 3.15T$
$Au_{36}Nd_{17}$ phase, format $(Au)_{0.679}(Nd)_{0.321}$
${}^{0}G^{Au_{36}Nd_{17}}_{Au:Nd} - 0.679{}^{0}G^{fcc}_{Au} - 0.321{}^{0}G^{dhcp}_{Nd} = -67500 - 5.332T$
Au_4Nd_3 phase, format $(Au)_{0.571}(Nd)_{0.429}$
${}^{0}G^{Au_4Nd_3}_{Au:Nd} - 0.571{}^{0}G^{fcc}_{Au} - 0.429{}^{0}G^{dhcp}_{Nd} = -74650 - 2.9496T$
γAuNd phase, format $(Au)_{0.5}(Nd)_{0.5}$
${}^{0}G^{AuNd}_{Au:Nd} - 0.5{}^{0}G^{fcc}_{Au} - 0.5{}^{0}G^{dhcp}_{Nd} = -76900 - 1.5076T$
βAuNd phase, format $(Au)_{0.5}(Nd)_{0.5}$
${}^{0}G^{AuNd}_{Au:Nd} - 0.5{}^{0}G^{fcc}_{Au} - 0.5{}^{0}G^{dhcp}_{Nd} = -79000 - 0.2224T$
αAuNd phase, format $(Au)_{0.5}(Nd)_{0.5}$
${}^{0}G^{AuNd}_{Au:Nd} - 0.5{}^{0}G^{fcc}_{Au} - 0.5{}^{0}G^{dhcp}_{Nd} = -79100 - 0.0802T$
$AuNd_2$ phase, format $(Au)_{0.333}(Nd)_{0.667}$
${}^{0}G^{AuNd_2}_{Au:Nd} - 0.333{}^{0}G^{fcc}_{Au} - 0.667{}^{0}G^{dhcp}_{Nd} = -53800 - 0.709T$

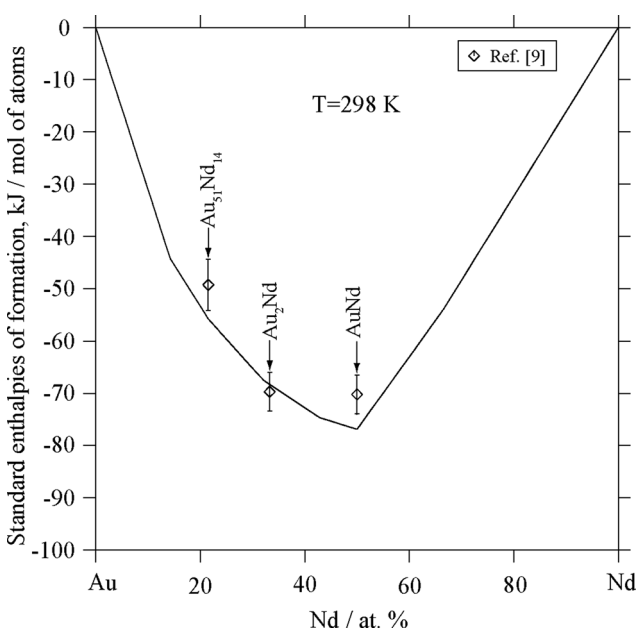

Fig. 2 Calculated standard enthalpies of formation at 298 K with reference states of Au (fcc) and αNd (dhcp) compared with the experimental data[9]

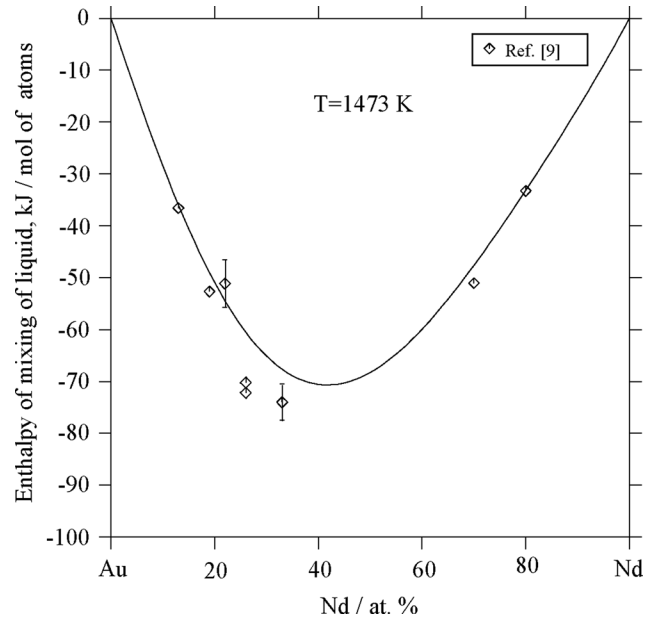

Fig. 3 Calculated enthalpies of mixing of liquid phase at 1473 K with reference states of pure elements of Au (liquid) and Nd (liquid) compared with the experimental data[9]

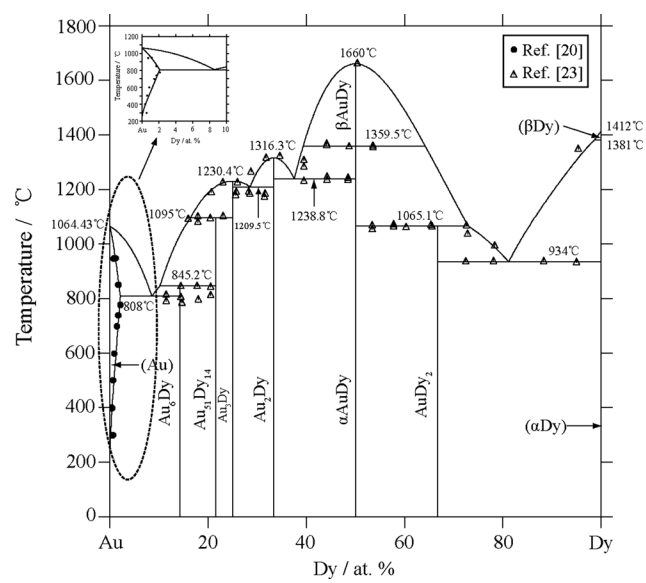

Fig. 4 Calculated phase diagram of the Au-Dy system compared with the experimental data[20,23]

are given in Fig. 6. It is seen that the calculated standard enthalpy of formation of the Au_2Dy phase and enthalpies of mixing are slightly more positive than the experimental value[24] and predicted data,[25] respectively. By considering the agreements of the most of experimental data including the data of phase equilibria and thermodynamic properties, this calculated result can be accepted.

Table 3 Comparison between the calculated and experimental results[20,23] in the Au-Dy system

Invariant reaction	Reaction type	Composition at. % Dy			Temperature, °C	Reference
$L \leftrightarrow \alpha Dy + AuDy_2$	Eutectic	79.0	···	66.7	935	[23]
		81.2	100	66.7	934	This work
$L + \alpha AuDy \leftrightarrow AuDy_2$	Peritectic	···	50.0	66.7	1065	[23]
		72.9	50.0	66.7	1065.1	This work
$L \leftrightarrow \beta AuDy$	Congruent		50.0		1660	[23]
			50.0		1660	This work
$\beta AuDy \leftrightarrow \alpha AuDy$	Allotropic		50.0		1360	[23]
			50.0		1359.5	This work
$L \leftrightarrow \alpha AuDy + Au_2Dy$	Eutectic	38	50.0	33.3	1235	[23]
		37.6	50.0	33.3	1238.8	This work
$L \leftrightarrow Au_2Dy$	Congruent		33.3		1320	[23]
			33.3		1316.3	This work
$L \leftrightarrow Au_2Dy + Au_3Dy$	Eutectic	27	33.3	25.0	1185	[23]
		28.6	33.3	25.0	1209.5	This work
$L \leftrightarrow Au_3Dy$	Congruent		25.0		1230	[23]
			25.0		1230.4	This work
$L + Au_3Dy \leftrightarrow Au_{51}Dy_{14}$	Eutectic	···	25.0	21.5	1095	[23]
		16.4	25.0	21.5	1095	This work
$L + Au_{51}Dy_{14} \leftrightarrow Au_6Dy$	Peritectic	···	21.5	14.3	845	[23]
		10.2	21.5	14.3	845.2	This work
$L \leftrightarrow Au_6Dy + (Au)$	Eutectic	···	···	2.3	808	[20]
		9.5	14.3	2.3	808	[23]
		8.5	14.3	2.1	808	This work

Table 4 Thermodynamic parameters in the Au-Dy system assessed in the present work

Parameters in each phase, J/mol
Liquid phase, format (Au, Dy)
${}^{0}L^{Liq}_{Au,Dy} = -273000 + 9.023T$
${}^{1}L^{Liq}_{Au,Dy} = -108900 - 7.863T$
${}^{2}L^{Liq}_{Au,Dy} = 15800 + 1.8T$
${}^{3}L^{Liq}_{Au,Dy} = 52800 - 3.6T$
Fcc phase, format (Au, Dy)
${}^{1}L^{Fcc}_{Au,Dy} = -292970$
Au_6Dy phase, format $(Au)_{0.857}(Dy)_{0.143}$
${}^{0}G^{Au_6Dy}_{Au:Dy} - 0.857{}^{0}G^{fcc}_{Au} - 0.143{}^{0}G^{hcp}_{Dy} = -52600 - 5.272T + 3.9TLn(T) - 0.0137T^2 + 2.66 * 10^{-6}T^3$
$Au_{51}Dy_{14}$ phase, format $(Au)_{0.785}(Dy)_{0.215}$
${}^{0}G^{Au_{51}Dy_{14}}_{Au:Dy} - 0.785{}^{0}G^{fcc}_{Au} - 0.215{}^{0}G^{hcp}_{Dy} = -61120 - 7.082T + 1.2TLn(T) - 0.0044T^2 + 2.86 * 10^{-6}T^3$
Au_3Dy phase, format $(Au)_{0.75}(Dy)_{0.25}$
${}^{0}G^{Au_3Dy}_{Au:Dy} - 0.75{}^{0}G^{fcc}_{Au} - 0.25{}^{0}G^{hcp}_{Dy} = -64000 - 4.93T + 1.09TLn(T) - 0.0163T^2 + 8.97 * 10^{-6}T^3$
Au_2Dy phase, format $(Au)_{0.667}(Dy)_{0.333}$
${}^{0}G^{Au_2Dy}_{Au:Dy} - 0.667{}^{0}G^{fcc}_{Au} - 0.333{}^{0}G^{hcp}_{Dy} = -69800 - 4.45T + 0.131TLn(T)$
βAuDy phase, format $(Au)_{0.5}(Dy)_{0.5}$
${}^{0}G^{AuDy}_{Au:Dy} - 0.5{}^{0}G^{fcc}_{Au} - 0.5{}^{0}G^{hcp}_{Dy} = -80963 + 0.98T$
αAuDy phase, format $(Au)_{0.5}(Dy)_{0.5}$
${}^{0}G^{AuDy}_{Au:Dy} - 0.5{}^{0}G^{fcc}_{Au} - 0.5{}^{0}G^{hcp}_{Dy} = -81363 + 1.225T$
$AuDy_2$ phase, format $(Au)_{0.333}(Dy)_{0.667}$
${}^{0}G^{AuDy_2}_{Au:Dy} - 0.333{}^{0}G^{fcc}_{Au} - 0.667{}^{0}G^{hcp}_{Dy} = -54300 - 0.215T$

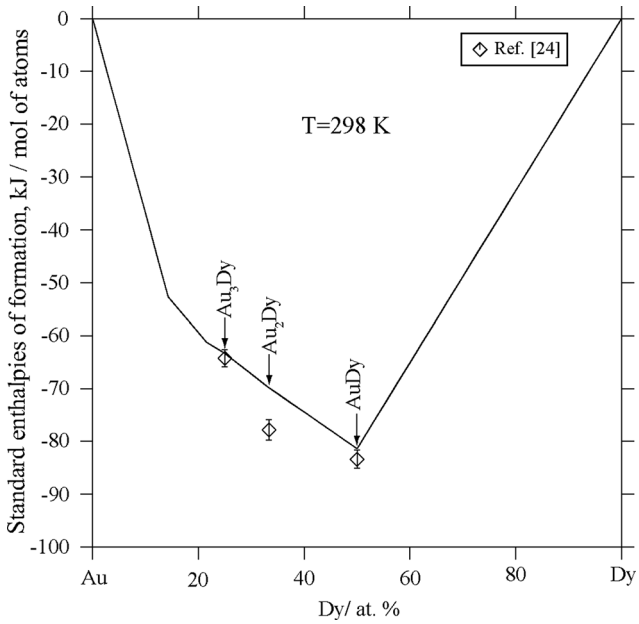

Fig. 5 Calculated standard enthalpies of formation at 298 K with reference states of Au (fcc) and αDy (hcp) compared with the experimental data[24]

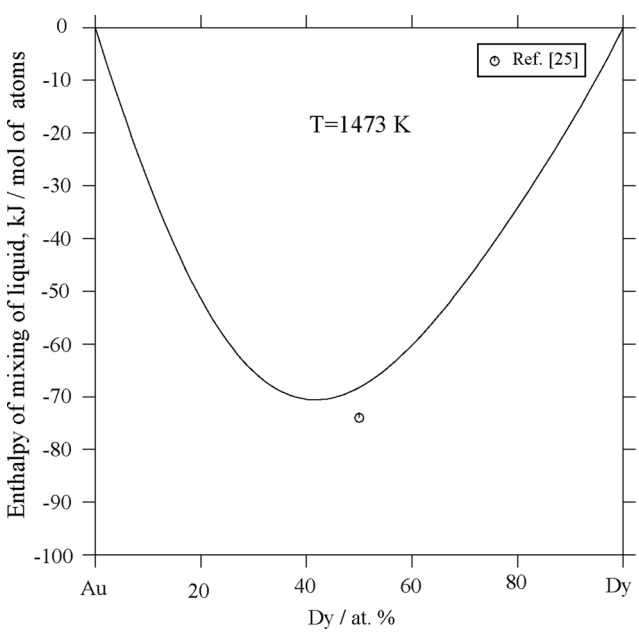

Fig. 6 Calculated enthalpies of mixing of liquid phase at 1473 K with reference states of pure elements of Au (liquid) and Dy (liquid) compared with the predicted data[25]

4. Conclusions

On the basis of available experimental data, the Au-RE (RE = Dy, Nd) binary systems have been critically optimized by the CALPHAD method. A consistent set of thermodynamic parameters has been obtained and most experimental information can be satisfactorily reproduced based on the assessed thermodynamic parameters.

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